GREEEN tallgrass prairie restoration: Seeded richness ~ Species pool size (NW Station)

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> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```r
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
library(lme4)
library(glmmTMB)
library(DHARMa)
library(emmeans)
```

Packages

```r
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    id_plot = "f",
    year = "f",
    seeding_time = col_factor(
      levels = c("unseeded", "fall", "spring"), ordered = FALSE
      ),
    seeded_pool = col_factor(
      levels = c("0", "6", "12", "18", "33"), ordered = TRUE
      ),
    treatment_id = "f",
    richness_type = "f"
  )
) %>%
  filter(
    site == "NW Station",
    richness_type == "seeded_richness",
    !(treatment_id %in% c("2", "4"))
  ) %>%
  mutate(y = richness_1qm + richness_25qm)
```

Load data

Statistics

Data exploration

Means and deviations

```r
Rmisc::CI(sites$y, ci = .95)
```

```
##    upper     mean    lower
## 5.025535 4.828125 4.630715
```

```r
median(sites$y)
```

```
## [1] 5
```

```r
sd(sites$y)
```

```
## [1] 1.386789
```

```r
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##  5% 95%
##   3   7
```

```r
sites %>% count(site, year)
```

```
## # A tibble: 4 x 3
##   site       year      n
##   <chr>      <fct> <int>
## 1 NW Station 2015     48
## 2 NW Station 2016     48
## 3 NW Station 2017     48
## 4 NW Station 2018     48
```

```r
sites %>% count(seeded_pool)
```

```
## # A tibble: 4 x 2
##   seeded_pool     n
##   <ord>       <int>
## 1 6              48
## 2 12             48
## 3 18             48
## 4 33             48
```

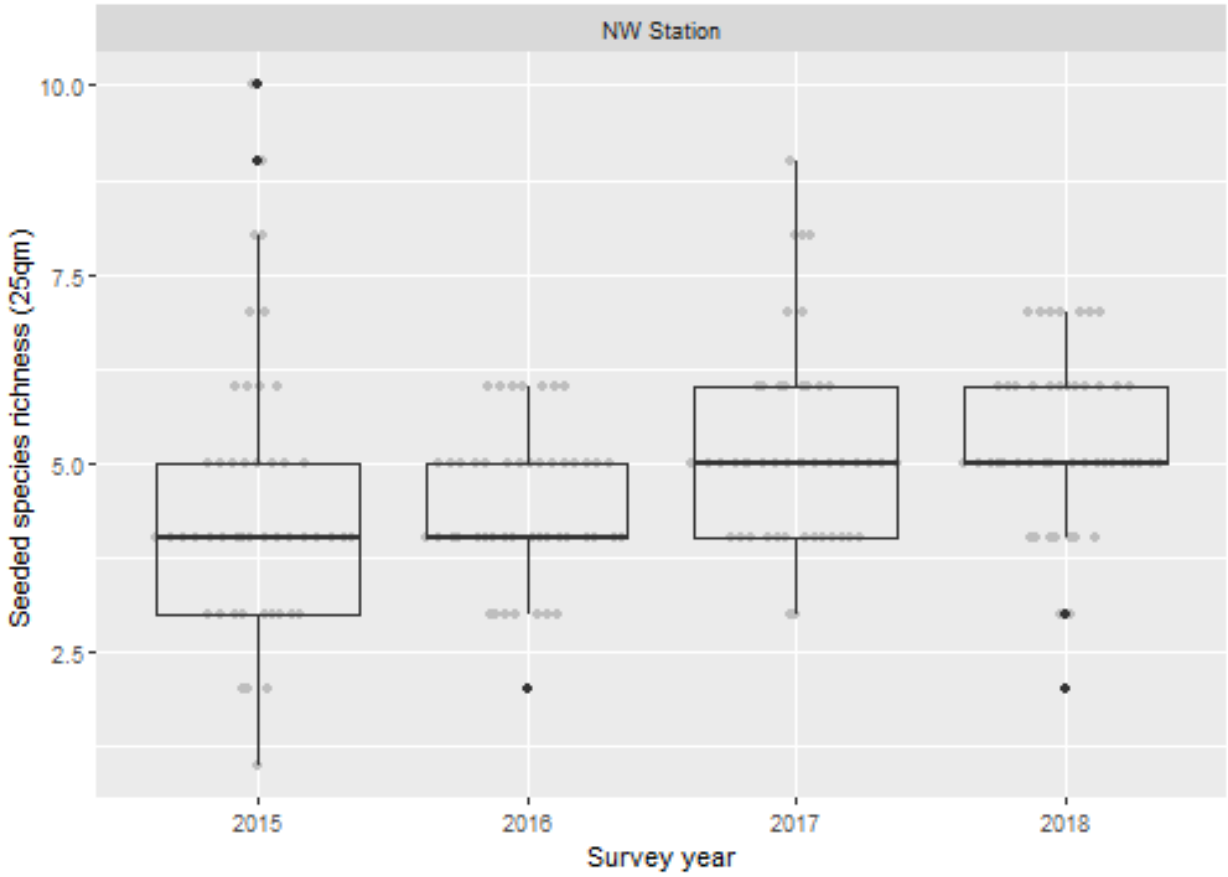
```r
sites %>% count(seeding_time)
```

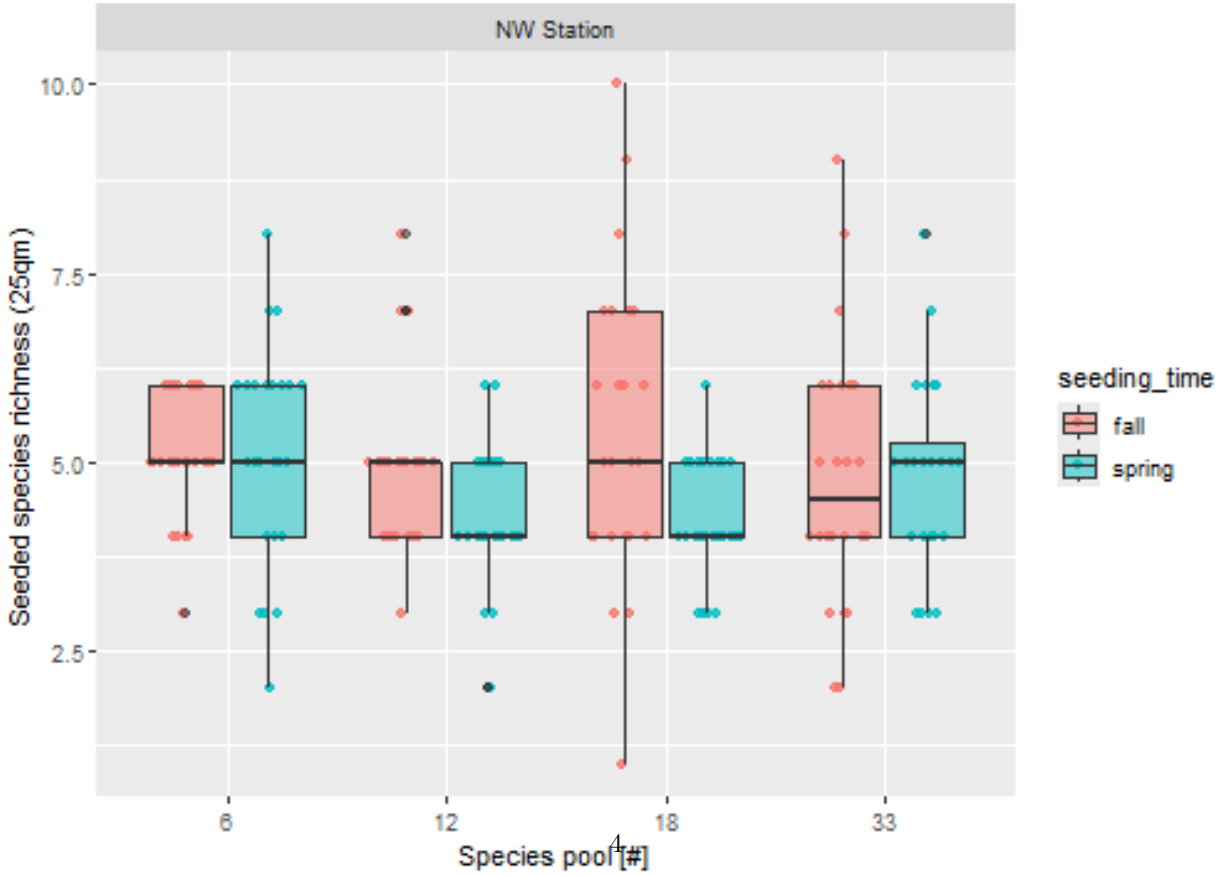
```
## # A tibble: 2 x 2
##   seeding_time     n
##   <fct>        <int>
## 1 fall            96
## 2 spring          96
```

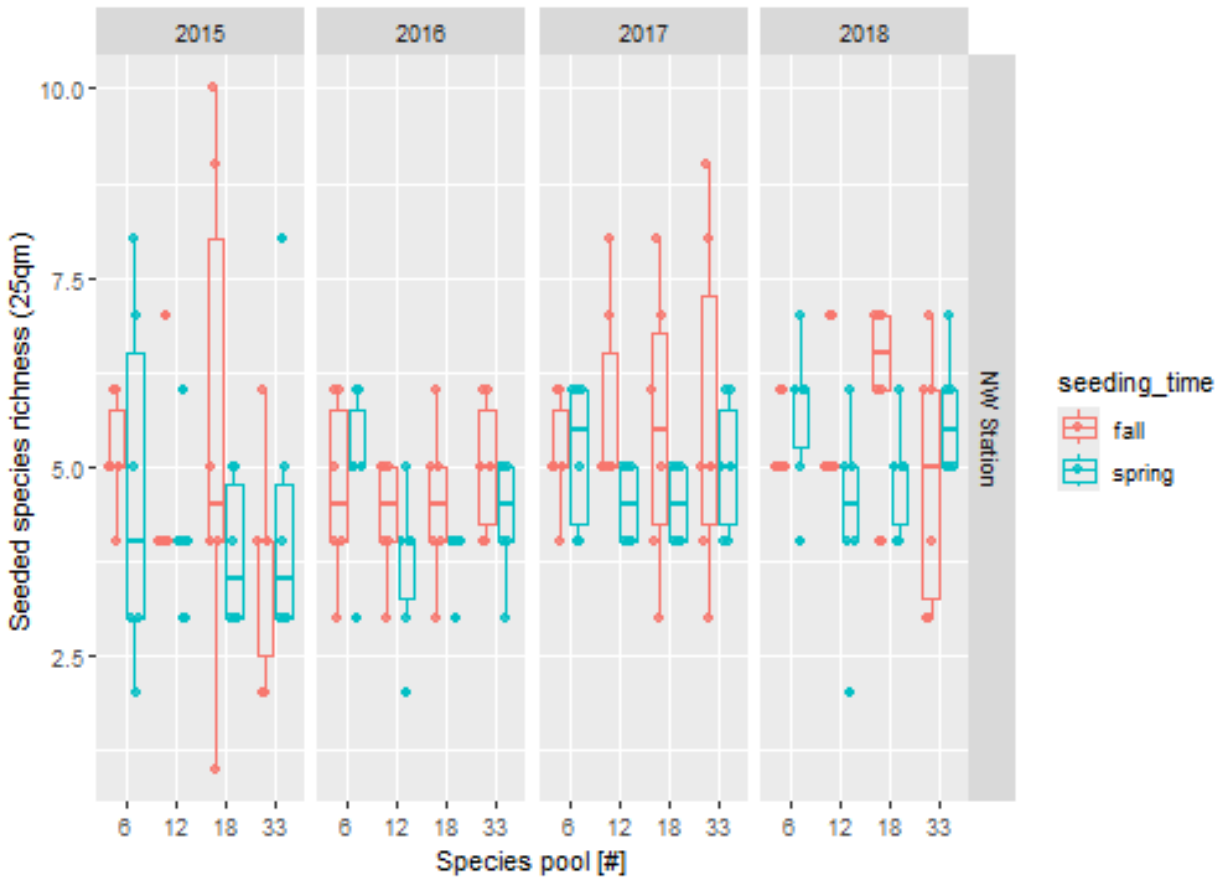
```r
sites %>% count(herbicide)
```

```
## # A tibble: 2 x 2
##   herbicide     n
##       <dbl> <int>
## 1         0    96
## 2         1    96
```

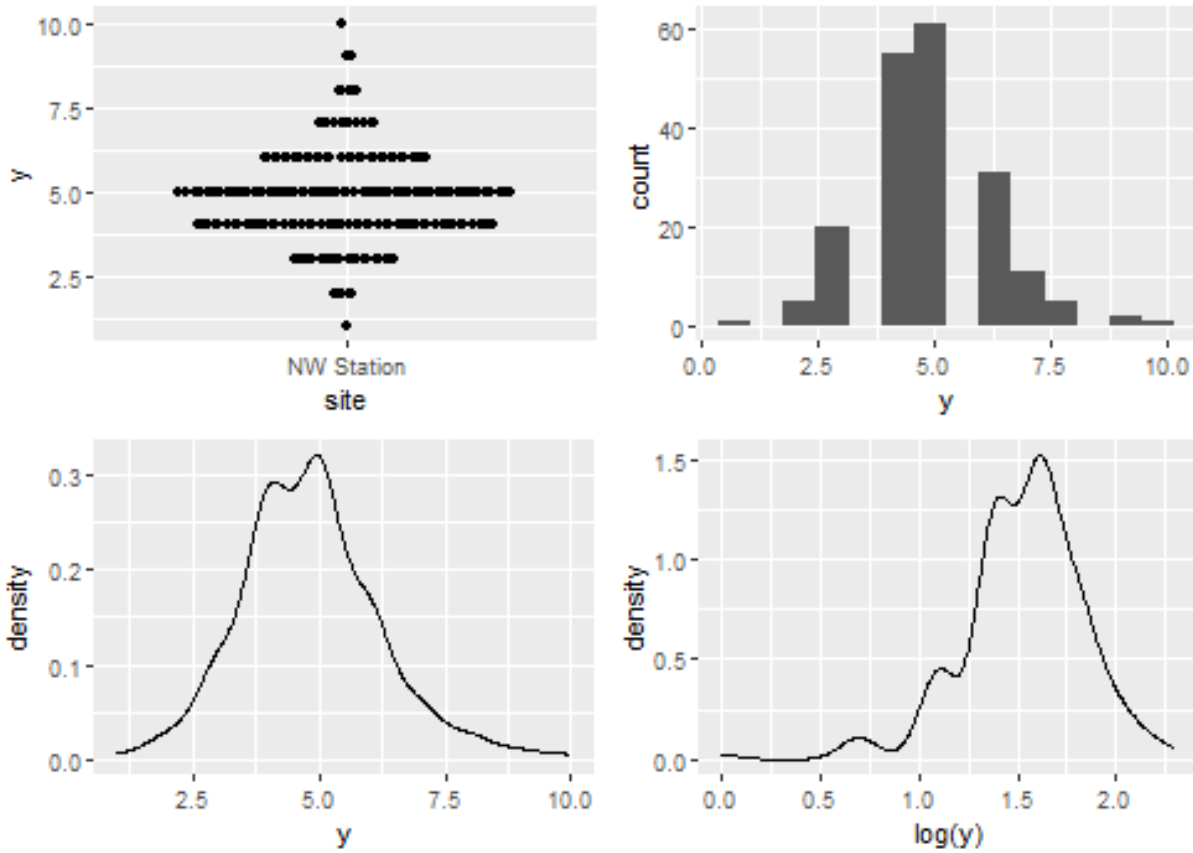
Graphs of raw data (Step 2, 6, 7)






2015
2016
2017
2018
NW Station
Seeded species richness (25qm)
10.0
7.5
5.0
2.5
6
12
18
33
Species pool [#]
seeding_time
fall
spring

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

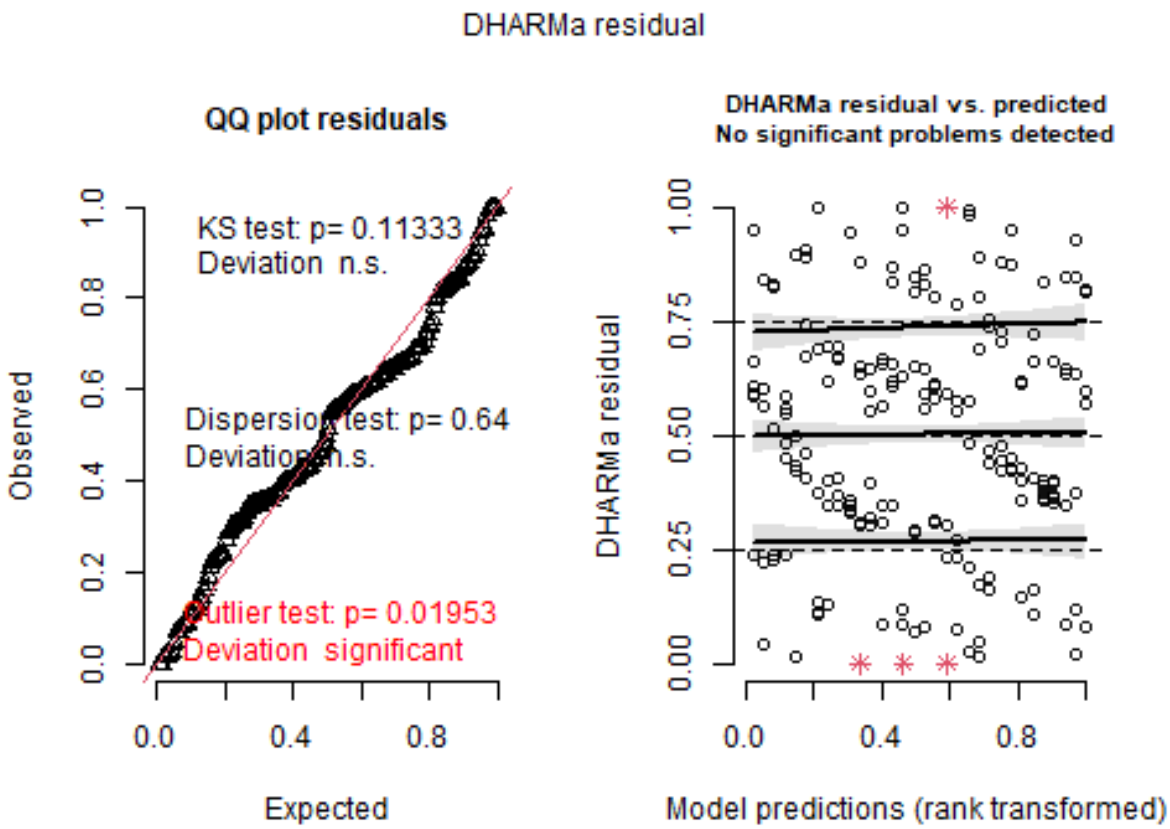
```
load(file = here("outputs", "models", "model_richness_pool_nws_3.Rdata"))
m_1 <- m3
```

```
formula(m_1)
## sqrt(y) ~ seeded_pool * seeding_time + year
```

Model check

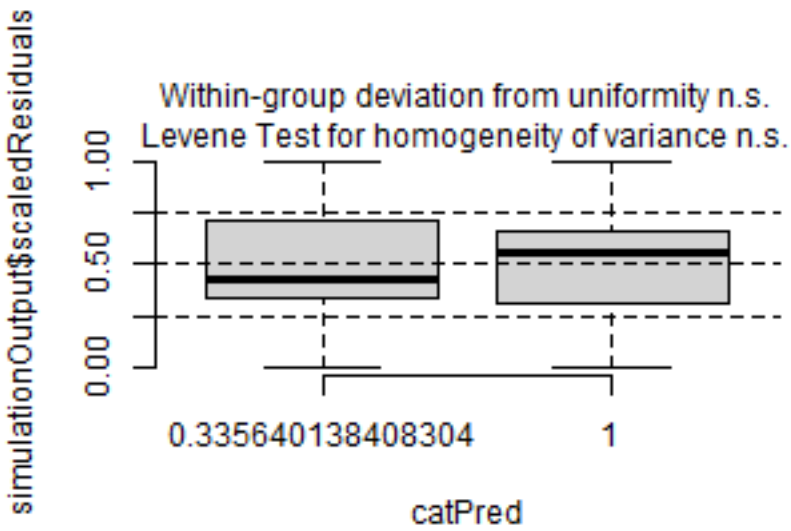
DHARMa

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```

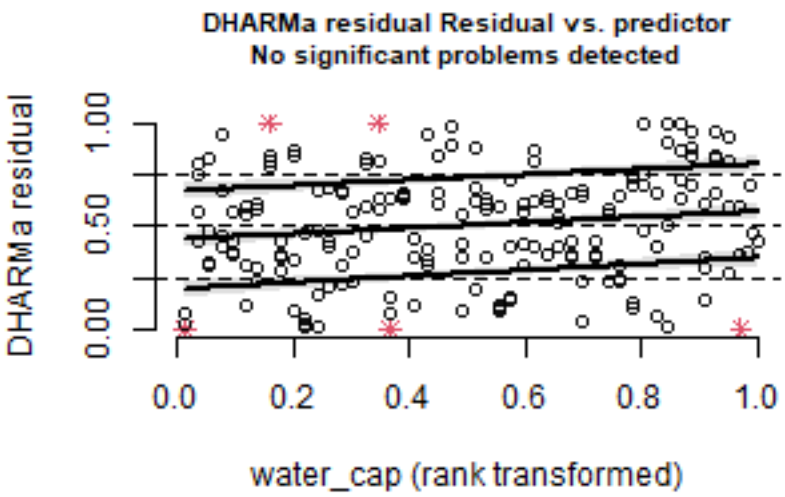



```
# simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

```
plotResiduals(simulation_output_1$scaledResiduals, sites$herbicide)
```




```
# plotResiduals(simulation_output_2$scaledResiduals, sites$herbicide)
# plotResiduals(simulation_output_1$scaledResiduals, sites$seeding_time)
# plotResiduals(simulation_output_2$scaledResiduals, sites$seeding_time)
# plotResiduals(simulation_output_1$scaledResiduals, sites$year)
# plotResiduals(simulation_output_2$scaledResiduals, sites$year)
plotResiduals(simulation_output_1$scaledResiduals, sites$water_cap)
```




```
# plotResiduals(simulation_output_2$scaledResiduals, sites$water_cap)
```

Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

```
##                           GVIF Df GVIF^(1/(2*Df))
## seeded_pool                  8  3        1.414214
## seeding_time                 1  1        1.000000
## year                         1  3        1.000000
## seeded_pool:seeding_time     8  3        1.414214
```

```
# car::vif(m_2)
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##            R2m       R2c
## [1,] 0.1523651 0.1523651
# MuMIn::r.squaredGLMM(m_2)
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
# MuMIn::AICc(m_1, m_2) %>%
#   arrange(AICc)
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)
```

```
## Anova Table (Type III tests)
##
## Response: sqrt(y)
##                           Sum Sq  Df   F value    Pr(>F)
## (Intercept)              173.471   1 1929.9817 < 2.2e-16 ***
## seeded_pool                0.205   3    0.7613 0.5171596
## seeding_time               0.477   1    5.3077 0.0223675 *
## year                       1.653   3    6.1316 0.0005402 ***
## seeded_pool:seeding_time   0.629   3    2.3340 0.0755229 .
## Residuals                 16.269 181
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

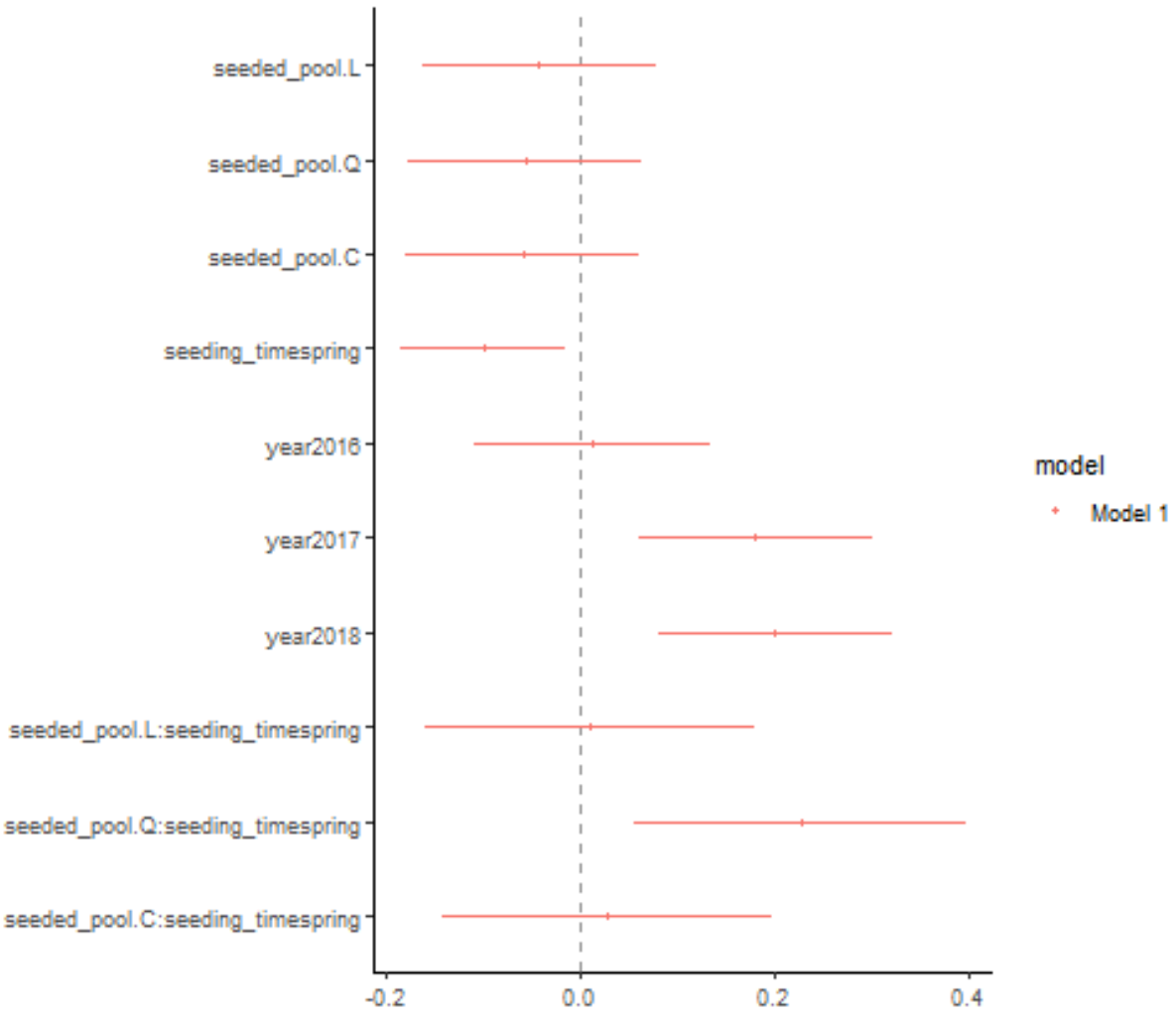
```
summary(m_1)
```

```
##
## Call:
## lm(formula = sqrt(y) ~ seeded_pool * seeding_time + year, data = sites)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.18385 -0.13523  0.03344  0.12905  0.97843
##
## Coefficients:
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      2.125435   0.048381  43.932  < 2e-16 ***
```

```
## seeded_pool.L                    -0.042979   0.061197  -0.702  0.48340
## seeded_pool.Q                    -0.056434   0.061197  -0.922  0.35767
## seeded_pool.C                    -0.059342   0.061197  -0.970  0.33349
## seeding_timespring               -0.099694   0.043273  -2.304  0.02237 *
## year2016                          0.012454   0.061197   0.204  0.83897
## year2017                          0.181122   0.061197   2.960  0.00349 **
## year2018                          0.201031   0.061197   3.285  0.00122 **
## seeded_pool.L:seeding_timespring  0.009487   0.086546   0.110  0.91283
## seeded_pool.Q:seeding_timespring  0.227130   0.086546   2.624  0.00942 **
## seeded_pool.C:seeding_timespring  0.027733   0.086546   0.320  0.74900
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2998 on 181 degrees of freedom
## Multiple R-squared:  0.1594, Adjusted R-squared:  0.113
## F-statistic: 3.433 on 10 and 181 DF,  p-value: 0.0003826
```

Forest plot

```r
dotwhisker::dwplot(
  list(m_1),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```



Effect sizes

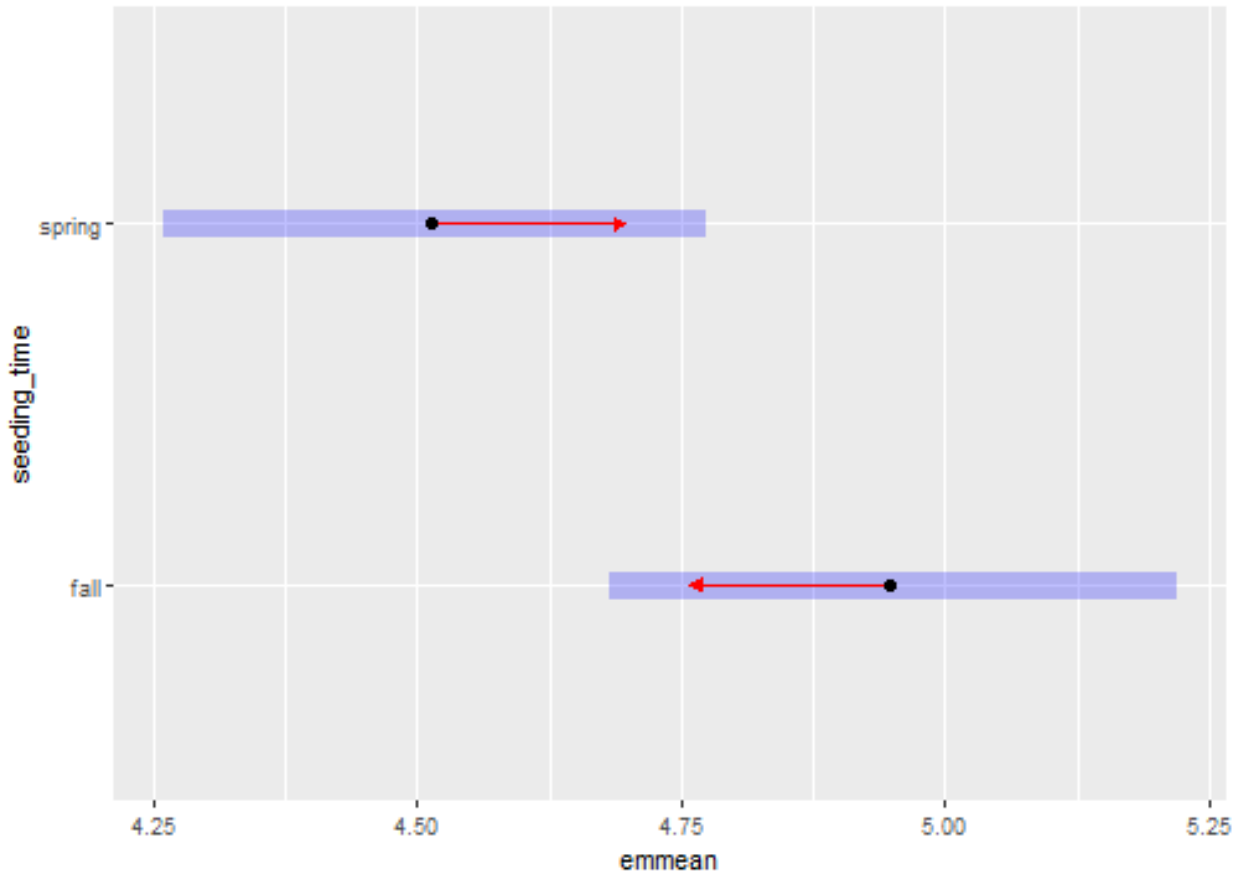
Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(
  m_1,
  revpairwise ~ seeding_time,
  type = "response"
  ))
```

```
## $emmeans
##  seeding_time response    SE  df lower.CL upper.CL
##  fall             4.95 0.136 181     4.68     5.22
##  spring           4.51 0.130 181     4.26     4.77
##
## Results are averaged over the levels of: seeded_pool, year
## Confidence level used: 0.95
## Intervals are back-transformed from the sqrt scale
##
## $contrasts
```

```
##  contrast      estimate     SE  df t.ratio p.value
##  spring - fall  -0.0997 0.0433 181  -2.304  0.0224
##
## Results are averaged over the levels of: seeded_pool, year
## Note: contrasts are still on the sqrt scale. Consider using
##       regrid() if you want contrasts of back-transformed estimates.
```

```r
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
```

```
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.11.1   DHARMa_0.4.7     glmmTMB_1.1.12   lme4_1.1-37
##  [5] Matrix_1.7-3     patchwork_1.3.1  ggbeeswarm_0.7.2 lubridate_1.9.4
##  [9] forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4      purrr_1.1.0
## [13] readr_2.1.5      tidyr_1.3.1      tibble_3.3.0     ggplot2_3.5.2
## [17] tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.4           gridExtra_2.3          sandwich_3.1-1
##  [4] rlang_1.1.6            magrittr_2.0.3         compiler_4.5.0
##  [7] mgcv_1.9-3             vctrs_0.6.5            pkgconfig_2.0.3
## [10] crayon_1.5.3           fastmap_1.2.0          backports_1.5.0
## [13] labeling_0.4.3         utf8_1.2.6             ggstance_0.3.7
## [16] promises_1.3.3         rmarkdown_2.29         tzdb_0.5.0
## [19] nloptr_2.2.1           bit_4.6.0              xfun_0.52
## [22] later_1.4.2            parallel_4.5.0         R6_2.6.1
## [25] gap.datasets_0.0.6     stringi_1.8.7          qgam_2.0.0
## [28] RColorBrewer_1.1-3     car_3.1-3              boot_1.3-31
## [31] numDeriv_2016.8-1.1    estimability_1.5.1     Rcpp_1.1.0
## [34] iterators_1.0.14       knitr_1.50             zoo_1.8-14
## [37] parameters_0.27.0      httpuv_1.6.16          splines_4.5.0
## [40] timechange_0.3.0       tidyselect_1.2.1       rstudioapi_0.17.1
## [43] abind_1.4-8            yaml_2.3.10            MuMIn_1.48.11
## [46] doParallel_1.0.17      TMB_1.9.17             codetools_0.2-20
## [49] lattice_0.22-7         plyr_1.8.9             shiny_1.11.1
## [52] withr_3.0.2            bayestestR_0.16.1      coda_0.19-4.1
## [55] evaluate_1.0.4         marginaleffects_0.28.0 pillar_1.11.0
## [58] gap_1.6                carData_3.0-5          foreach_1.5.2
## [61] stats4_4.5.0           reformulas_0.4.1       insight_1.3.1
## [64] generics_0.1.4         vroom_1.6.5            rprojroot_2.0.4
## [67] hms_1.1.3              scales_1.4.0           minqa_1.2.8
## [70] xtable_1.8-4           glue_1.8.0             tools_4.5.0
## [73] data.table_1.17.8      mvtnorm_1.3-3          grid_4.5.0
## [76] rbibutils_2.3          datawizard_1.1.0       nlme_3.1-168
## [79] Rmisc_1.5.1            performance_0.15.0     beeswarm_0.4.0
## [82] vipor_0.4.7            Formula_1.2-5          cli_3.6.5
## [85] gtable_0.3.6           digest_0.6.37          farver_2.1.2
## [88] htmltools_0.5.8.1      lifecycle_1.0.4        mime_0.13
## [91] bit64_4.6.0-1          dotwhisker_0.8.4       MASS_7.3-65
```